

Stage 5: Extended Dimensionality (Higher Closures)

5.1 New Closure Operators – Definitions & Rationale

We introduce five new closure operators beyond the original four, each targeting a distinct structural aspect of complex systems. For each operator, we give a formal definition (an equivalence relation on states) and an intuitive explanation of what complexity it reduces, along with examples. We then show that each is a **closure operator** (satisfying idempotence, extensiveness, and reflectivity), meaning it can be applied without adding new distinctions and can be composed with the others.

Symmetry Closure ($\text{\textbackslash varsigma}_{\text{sym}}$)

Definition 5.1 (Symmetry Equivalence): Let G be a symmetry group (e.g. a Lie group or permutation group) acting on the state space $|X|$ in a way that preserves the system's metrics and dynamics. We declare two states equivalent if one is related to the other by a symmetry transformation ¹:

$$x_1 \sim_{\text{sym}} x_2 \iff \exists g \in G \text{ such that } x_2 = g \cdot x_1.$$

In other words, x_1 and x_2 lie in the same G -orbit. The **symmetry closure** $\text{\textbackslash varsigma}_{\text{sym}}(X)$ then **quotients** the system by this relation, identifying each orbit $[x] = \{g \cdot x : g \in G\}$ as a single abstract state ¹. The resulting quotient system has state space $|X|/G$ (the set of orbits), with induced structures: distances are defined by the shortest distance between orbits, measures are pushed forward by the quotient map, and transition kernels average over group actions (e.g. integrating K_X over $g \in G$) ².

(Closure property): $\text{\textbackslash varsigma}_{\text{sym}}$ is indeed an idempotent **closure operator** (the orbit quotient is a **reflector** onto the subcategory of G -invariant systems) ³. Intuitively, applying the symmetry closure twice has no further effect, and the quotient $X \mapsto \text{\textbackslash varsigma}_{\text{sym}}(X)$ is an inclusion in the categorical sense (no new states are created, only identifications of symmetric ones).

Intuition & Examples: This closure eliminates redundant degrees of freedom arising from symmetry. If two configurations differ only by a symmetry transformation, they are physically or observationally indistinguishable, so $\text{\textbackslash varsigma}_{\text{sym}}$ merges them into one state. For example, in a gauge theory, quotienting by the gauge group G collapses states related by gauge transformations, yielding the physical state space as $|X|/G$ ⁴. In particle dynamics, if particles are identical, permuting them should not create distinct states; $\text{\textbackslash varsigma}_{\text{sym}}$ with $G = S_n$ (the permutation group) removes such spurious distinctions ⁴. By focusing on *invariant structure*, symmetry closure ensures the model distinguishes only what is physically meaningful.

Causal Closure (Σ_{caus})

Definition 5.2 (Causal Equivalence): Two states x_1, x_2 are defined to be *causally equivalent* if they have the same influence on the past and future of the system, meaning no experiment or intervention can tell them apart in terms of outcomes. Formally, we require that ⁵:

- **Arrival histories:** x_1 and x_2 have identical distributions of arrival times from any source or past condition (i.e. any way the system could evolve into those states).
- **Future responses:** For every possible intervention or context, the conditional distribution of future trajectories (or next states) from x_1 is the same as from x_2 . Equivalently, they have the same Markov transition probabilities and outcome statistics under all tests.

If $x_1 \sim_{\text{caus}} x_2$, then $\Sigma_{\text{caus}}(X)$ merges x_1 and x_2 into a single state. The **causal closure** $\Sigma_{\text{caus}}(X)$ is the quotient of X by this relation, so that each equivalence class represents a *causally indivisible* state (no hidden differences in how they got there or where they will lead) ⁵.

(Closure property): Σ_{caus} is an idempotent closure operator. Applying it compresses the state space without losing any causal information: in fact, the reduced system $\Sigma_{\text{caus}}(X)$ still generates the same set of possible influences and responses as X did, just on fewer states ⁶. One can show that the reduced Markov kernel K_{caus} on $\Sigma_{\text{caus}}(X)$ yields a Markov process that is **causal-equivalent** to the original, in the sense of preserving all probabilities of future events given past histories (a concept related to Pearl’s causal calculus of *sufficient statistics*) ⁶.

Intuition & Examples: This closure eliminates state distinctions that have no effect on future dynamics or that cannot be distinguished by any causal intervention. In essence, if two states “behave the same” in terms of how the system can evolve and respond, they are treated as one. For example, in epidemiology, two patients might have different exposure histories but if they contribute identically to disease spread going forward, Σ_{caus} will merge them ⁷. In a reinforcement learning context, two agent states that yield identical outcomes for all possible action sequences are causally equivalent. By applying causal closure, we simplify the system to only those distinctions that influence future behavior, removing non-Markovian “memory” distinctions that don’t matter for prediction.

Contextual Closure (Σ_{ctx})

Definition 5.3 (Contextual Equivalence): Many systems behave differently under varying external conditions or parameters (contexts). We say two states x_1, x_2 are *context-equivalent* if **no context can tell them apart** – that is, when we average over any possible environmental or parametric context, the two states have identical behavior. Formally, assume the system’s dynamics K_X^θ can depend on an external context $\theta \in \Theta$, with θ drawn from some distribution ξ . We require that for every choice of context distribution ξ , the expected transition probabilities from x_1 equal those from x_2 ⁸:

$$x_1 \sim_{\text{ctx}} x_2 \iff \forall \text{ distributions } \xi: \mathbb{E}_{\theta \sim \xi}[K_X^\theta(x_1, \cdot)] = \mathbb{E}_{\theta \sim \xi}[K_X^\theta(x_2, \cdot)].$$

In simpler terms, under any fixed contextual conditions, x_1 and x_2 have the same probabilistic outcomes; hence when one “integrates out” context, they are indistinguishable. The **contextual closure** Σ_{ctx}

$\text{varsigma}_{\text{ctx}}(X)$ then identifies such states, yielding a quotient system where each equivalence class represents a state that is robust across all contexts ⁸.

(*Closure property*): $\text{varsigma}_{\text{ctx}}$ is a closure operator (idempotent and extensive). The quotient $\text{varsigma}_{\text{ctx}}(X)$ produces a reduced model that is **robust to context variations**: by construction, the behavior of each merged state is invariant under changes in context distribution ⁹. No further identifications can be made without altering this property, and applying the closure twice has no additional effect.

Intuition & Examples: Contextual closure merges states that differ only in ways that are relevant to specific contexts but *irrelevant on average*. This is analogous to identifying phenotypically similar behaviors that persist across different environments. For instance, in a gene regulatory network, two molecular states might behave identically when averaged over various environmental conditions (temperature, nutrients); $\text{varsigma}_{\text{ctx}}$ would merge such states, reflecting that the distinction doesn't matter unless a specific context is singled out ¹⁰. In control systems, $\text{varsigma}_{\text{ctx}}$ will identify states that respond the same way across a range of uncertainties or operating conditions, thus “forgetting” context-specific quirks and retaining only robust behaviors. Conceptually, this closure is like a **sheaf-like gluing** of states across all contexts: if two states cannot be distinguished by *any* context, they are considered the same state in the quotient. This ensures no information that is purely context-dependent remains in the reduced model.

Cohomological Closure ($\text{varsigma}_{\text{coh}}$)

Definition 5.4 (Homological Equivalence): Complex systems often have important **topological** features in their state-space (think of cycles, holes, connectivity structures). We define an equivalence relation that identifies states lying in the same topological invariant. Concretely, consider the state space $|X|$ as a topological space (using, for example, the topology induced by the metric d_X). Compute its homology groups $H_k(|X|)$ for various dimensions k (these capture k -dimensional holes or cycles in the space). Say that a state x *supports* a homology class $h \in H_k$ if x lies on a cycle (or in a region) that represents h . Then we declare ¹¹:

$$x_1 \sim_{\text{coh}} x_2 \iff \text{for every homology class } h, x_1 \text{ supports } h \iff x_2 \text{ supports } h.$$

In other words, x_1 and x_2 belong to the same “homological sector” of the state space – they participate in the same topological features. The **cohomological closure** $\text{varsigma}_{\text{coh}}(X)$ collapses the state space accordingly: intuitively, states that differ only by wandering within the same topological loop or cavity are merged ¹². The quotient's states can be thought of as the distinct combinations of topological features present at those points.

(*Closure property*): $\text{varsigma}_{\text{coh}}$ is an idempotent closure operator. The reduced system $\text{varsigma}_{\text{coh}}(X)$ preserves the important topological invariants of the original. In fact, one can show that it maintains the full *persistent homology* information of X : any cycle or hole that existed in X corresponds to a (perhaps simplified) cycle or hole in $\text{varsigma}_{\text{coh}}(X)$ with the same persistence (appearance and disappearance scales in a filtered space) ¹³. Thus, $\text{varsigma}_{\text{coh}}$ “forgets” only topologically redundant information (like points along the same cycle) but not the existence of the cycle itself.

Intuition & Examples: Cohomological closure reduces the state-space by **contracting unimportant geometry while retaining fundamental topology**. For instance, consider a large network graph that has several loops or cycles: Σ_{coh} will merge nodes and edges in such a way that those loops are kept (perhaps as single representative cycles), but extra detours or parallel paths that don't create new homology might be collapsed. In topological data analysis, this is akin to simplifying a point cloud to its "topological skeleton" – points that form the same hole or cluster are identified, preserving the hole ¹⁴. In materials science, if a molecular configuration space has a persistent cavity or tunnel structure, cohomological closure will ensure that structure remains in the reduced model, while micro-level differences that do not affect connectivity (like slight deformations along a loop) are removed. Essentially, this closure targets **global structural invariants** (homology) and removes distinctions that are topologically trivial.

Trajectory Closure (Σ_{traj})

Definition 5.5 (Trajectory Equivalence): Instead of individual states, the trajectory closure considers entire *paths* through the state space. Let $\Pi(X)$ denote the set of all possible trajectories (infinite paths or finite-horizon sequences) that the system can take, given its dynamics K_X . We define an equivalence relation on $\Pi(X)$: two trajectories γ_1 and γ_2 are equivalent if they "look the same" up to a certain notion of path signature. For instance ¹⁵:

- They may be defined to have identical **signature** in the sense of rough path theory or time-series invariants (e.g. the same sequence of events or same integrals of motion along the path).
- Equivalently, one can require they are *rough-path equivalent*: γ_1 and γ_2 agree on all iterated integrals up to a certain order (so they induce the same response in any controlled differential equation up to a given complexity).

If $\gamma_1 \sim_{\text{traj}} \gamma_2$ by such a criterion, then the **trajectory closure** Σ_{traj} identifies these two trajectories as one abstract trajectory in the quotient. In practice, $\Sigma_{\text{traj}}(X)$ can be seen as working in a *path space category*, where we factor out trajectories that have the same overall effect ¹⁵.

(*Closure property*): Σ_{traj} is a closure operator (here formally defined on an enriched category of path-space objects). It is idempotent: merging equivalent trajectories twice does nothing beyond merging them once. The quotient $\Sigma_{\text{traj}}(X)$ can be thought of as a condensed representation of all system executions, where many different raw trajectories map to the same equivalence class if they are behaviorally the same ¹⁶. This reduced trajectory space preserves all *control-relevant* or *outcome-relevant* information from the original set of trajectories.

Intuition & Examples: Trajectory closure addresses temporal complexity by quotienting entire histories that lead to the same result. It recognizes that in many systems, different sequences of intermediate states might not actually matter – only the overall effect or end state does. For example, in a planning or control problem, there could be multiple distinct routes to get a robot arm from point A to point B; if those routes have identical cost and final configuration, Σ_{traj} would treat them as one trajectory. In reinforcement learning, two different action sequences that yield the same cumulative reward and state changes are effectively equivalent in terms of decision-making and can be merged. In stochastic processes, this relates to the idea of *coarse-graining path space*: trajectories that produce the same time-series of observable events can be identified. By compressing paths in this way, we drastically reduce the complexity

of analyzing protocols or behaviors, focusing on the essential *pattern of the trajectory* rather than the minutiae of how it is executed ¹⁷ .

Each of the above five closures targets a different structural aspect (symmetry, causality, context-dependence, topology, and trajectories). We have given their formal definitions and sketched why each satisfies the closure axioms (idempotence, etc.). In each case, the quotient can be understood as a **lossless reduction** along that particular dimension: it removes only the differences that are irrelevant to that aspect, while preserving all the information that is relevant. We next consider how these new closures interact with the original four (algebraic, geometric, spectral, logical) and with each other.

5.2 Unified Dimensionality Theorem – Commutativity and Idempotence

With nine closure operators $\{\sigma_a, \sigma_g, \sigma_s, \sigma_{\ell}, \sigma_{\text{sym}}, \sigma_{\text{caus}}, \sigma_{\text{ctx}}, \sigma_{\text{coh}}, \sigma_{\text{traj}}\}$ in our framework, an important question is whether we can apply all of them together *without conflict*. The **Unified Dimensionality Theorem** affirms that we can: all these closures commute with each other and can be composed in any order, yielding a single combined reduction that is itself a closure operator. We outline the key properties and their implications:

Pairwise Commutativity of Closures: Every pair of closure operators in the set commutes (in the categorical sense that $\sigma_i \circ \sigma_j$ is naturally isomorphic to $\sigma_j \circ \sigma_i$). In practical terms, applying one closure and then another yields the same result as applying them in the opposite order ¹⁸ . This pairwise commutativity has been established for each new closure with respect to each other and with respect to the original four. For example:

- **Algebraic vs. Symmetry:** Merging behaviorally equivalent states (algebraic closure) and then factoring out symmetry orbits yields the same quotient as doing symmetry first and then merging behaviors. Intuitively, group symmetries do not create new behavioral distinctions, and behaviorally indistinguishable states remain so under symmetry transformations ¹⁹ .
- **Spectral vs. Causal:** Reducing a Markov chain by slow eigenmodes (spectral closure) commutes with identifying causally equivalent states. Each operation preserves the Markovian structure in a way that doesn't interfere with the other – spectral aggregation doesn't break causal relationships, and quotienting causal structure doesn't alter the slow vs. fast modes ²⁰ .
- **Logical vs. Contextual:** If two states cannot be distinguished by any logical predicate or observation (logical closure), this remains true even when considering varying contexts – merging context-equivalent states doesn't introduce new observable differences. Conversely, two states equivalent under all context distributions will also be indistinguishable by any logical property that is robust to context. Thus, logical and contextual closures commute ²¹ .
- **Geometric vs. Cohomological:** Collapsing states that are close in metric space (geometric closure, which clusters small-distance states) does not eliminate large-scale topological features like loops (homological invariants), and vice versa. Whether we first cluster nearby points and then identify homologically similar regions, or do it in the opposite order, the end result is the same reduced space, because metric clustering (local) and homology identification (global) capture independent facets of the structure ²² .
- **Trajectory vs. Geometric:** Considering whole-path equivalences (trajectory closure) commutes with instantaneous-state merging based on geometry. Intuitively, if two trajectories are deemed equivalent based on their path signatures, this judgment isn't upset by additionally identifying states

that were close together (since truly equivalent trajectories will remain equivalent under slight state-space contractions). Likewise, merging nearby states first and then looking at trajectories yields the same outcome as identifying equivalent trajectories first – in both cases, consistency is maintained (one can formalize this via the continuity of the exponential map mapping local neighborhoods to global paths) ²³ .

These are just representative pairings; in fact **every** pair among the nine closures can be shown to commute naturally. The important consequence is that **the composition of all nine closures is order-independent**. No matter in which sequence we apply $\sigma_a, \sigma_g, \dots, \sigma_{\text{traj}}$, we will arrive at the same final quotient of X (up to isomorphism) ²⁴ . This justifies speaking of applying “all closures at once” — the end result does not depend on an ordering or interleaving of these reductions.

Global Aggregator $\mathcal{Q}_n$ – Definition: We now define the **extended global aggregator** $\mathcal{Q}_n$ as the composite of all nine closure operators. Starting from the original aggregate $\mathcal{Q} = \sigma_{\text{ell}} \circ \sigma_s \circ \sigma_g \circ \sigma_a$ (the four-aspect closure), we extend it as follows:

$$\mathcal{Q}_n := \sigma_{\text{traj}} \circ \sigma_{\text{coh}} \circ \sigma_{\text{ctx}} \circ \sigma_{\text{caus}} \circ \sigma_{\text{sym}} \circ \mathcal{Q},$$

in effect $\mathcal{Q}_n = \sigma_{\text{traj}} \circ \sigma_{\text{coh}} \circ \sigma_{\text{ctx}} \circ \sigma_{\text{caus}} \circ \sigma_{\text{sym}} \circ \sigma_{\text{ell}} \circ \sigma_s \circ \sigma_g \circ \sigma_a$. Thanks to commutativity, this definition does not depend on the particular order shown; any permutation of the nine operators yields an equivalent composite operator ²⁵ . Thus, $\mathcal{Q}_n$ is well-defined as “apply all closures.” By construction $\mathcal{Q}_n(X)$ is a system that is simultaneously closed under $\sigma_a, \sigma_g, \dots, \sigma_{\text{traj}}$ – i.e. it has no further reducible redundancy in any of these dimensions.

Idempotence of $\mathcal{Q}_n$: Perhaps the most crucial property of the composite reduction is that it remains a closure operator itself. In particular, applying all closures twice is the same as applying them once: $\mathcal{Q}_n^2 = \mathcal{Q}_n$. We can see this algebraically:

$$\mathcal{Q}_n^2 = (\sigma_1 \circ \sigma_2 \circ \dots \circ \sigma_n) \circ (\sigma_1 \circ \dots \circ \sigma_n).$$

Using commutativity, we can reorder this as $\sigma_1 \circ \sigma_1 \circ \sigma_2 \circ \sigma_2 \circ \dots \circ \sigma_n \circ \sigma_n$. Since each σ_i is individually idempotent ($\sigma_i^2 = \sigma_i$), this reduces to $\sigma_1 \circ \sigma_2 \circ \dots \circ \sigma_n$, which is $\mathcal{Q}_n$ again ²⁶ . More conceptually, after we have applied all the closures once (getting $\mathcal{Q}_n(X)$), the system is already σ_i -closed for each i , so applying σ_i again does nothing for each aspect. Idempotence implies $\mathcal{Q}_n(X)$ is a **fixed point** of $\mathcal{Q}_n$: it is the fully reduced version of X , and reducing it further yields no change. This property generalizes the Stage 4 result that $\mathcal{Q}$ (with four closures) was idempotent ²⁷ – now we have nine closures and still a one-step reduction to an irreducible form.

Lattice of Closure Combinations: The commutativity of all closures means that not only can we apply all of them together, but *any subset* of them can be applied in any order with a well-defined outcome. In fact, the set of all possible composite closures (applying various subsets of the σ_i ’s) has a rich structure: it forms a **lattice** isomorphic to the power set of $\{1, \dots, n\}$ ²⁸ . Each subset $S \subseteq \{1, \dots, 9\}$ corresponds to the composite operator $\mathcal{Q}_S := \prod \sigma_i$ (with the product taken in any

order). There is an inclusion order: if $S \subseteq T$, then applying the closures in T yields a *further reduced* system compared to applying those in S (since T has all the closures of S , possibly plus more). In the lattice, $S \leq T$ if and only if $S \subseteq T$ corresponds to the notion that $\mathcal{Q}_T(X)$ is “more reduced” (has identified at least as many distinctions) as $\mathcal{Q}_S(X)$. The lattice has:

- **Top element:** $S = \emptyset$ (no closures applied) which yields $\mathcal{Q}_\emptyset(X) = X$ itself (the original system, fully unreduced).
- **Bottom element:** $S = \{1, 2, \dots, n\}$ (all closures applied) which yields $\mathcal{Q}_{\{1, 2, \dots, n\}}(X) = \mathcal{Q}_n(X)$, the maximally reduced system.

Meet and join operations in this lattice correspond to intersections and unions of closure sets. For example, the meet of two closure sets S and T (their intersection $S \cap T$) corresponds to applying only the closures common to both – it yields a system that is *less reduced* (preserves more detail) than either of $\mathcal{Q}_S(X)$ or $\mathcal{Q}_T(X)$ individually, essentially reversing some reductions. Conversely, the join $S \cup T$ applies all closures from either set, yielding a system that is reduced along any aspect that either S or T would reduce. Because all the closures commute and are idempotent, this lattice behaves like a Boolean lattice of sub-equivalences²⁹: the order is just set inclusion. This result solidifies the idea that we have a *family* of consistent quotient models of X – ranging from the trivial identity (no reduction) down to the fully reduced $\mathcal{Q}_n(X)$ – and this family is structured and navigable (you can go down to a more reduced model by adding a closure, or back up by removing one).

Summarizing the **Unified Dimensionality Theorem**: if a suite of closures commute pairwise, then (1) they can be applied in any order or combination consistently, (2) the composite of all is an idempotent closure (a single-step global reduction), and (3) all combinations of them form a distributive lattice of models between X and $\mathcal{Q}_n(X)$. In our case, we have verified commutativity for the nine specific σ operators, so all these properties hold for the extended KUSMAN framework. This means **we can safely perform “all-aspects-at-once” reduction** and reason about partial reductions along different dimensions using the lattice structure.

5.3 Information Monotonicity and Optimality

An essential sanity check for adding new closures is that we should not be *gaining* any information by reducing a system – reductions should only remove or consolidate information. Indeed, each closure is designed as a kind of information compression (removing redundancies), and adding more such operations should not increase complexity. We formalize this intuition and show that the extended set of closures preserves the **information monotonicity** property: as we apply closures (individually or in combination), a suitable measure of information content never increases. Furthermore, we establish that $\mathcal{Q}_n(X)$ is the *minimal sufficient representation* of X with respect to the chosen aspects, in the sense that it contains no superfluous information but still retains all aspect-relevant information.

To quantify information, let $I(X)$ be a measure of the system’s information content or complexity. This could be, for example, the entropy of the distribution over states (if finite), the logarithm of the number of distinguishable states, or any other monotonic functional that reflects how much detail is in the system. We assume $I(\cdot)$ has the property that if one system’s state space is a *quotient* of another’s (a many-to-one identification), then its information content is less than or equal to the other’s.

Theorem 5.7 (Multidimensional Information Monotonicity and Optimality): For any system X in $\mathcal{MMK}$, let $X_{\text{red}} = \mathcal{Q}_n(X)$ be the fully reduced system obtained by applying all n closures. Then:

1. *Monotonicity:* $I(X_{\text{red}}) \leq I(X)$. In words, the information content of the reduced system is no greater than that of the original ³⁰. Moreover, $I(X_{\text{red}}) = I(X)$ if and only if X was already fully reduced (i.e. X was already invariant under all the closures, so $\mathcal{Q}_n(X) \cong X$). Every closure operation is information-non-increasing, so any composition of them is also information-non-increasing.
2. *Optimality:* $\mathcal{Q}_n(X)$ is the **minimal sufficient representation** of X with respect to the n aspects. This means that among all systems Y that preserve all the invariant information of X (all the algebraic, geometric, spectral, logical, symmetry, causal, contextual, topological, and trajectory-based distinctions that X had), the system $\mathcal{Q}_n(X)$ has the smallest state space (and in general the least complexity) ³¹. Formally, if Y is any other system such that there exists a morphism $X \rightarrow Y$ preserving all those structures (so Y is at least as coarse-grained as X along each aspect), then there is a morphism $\mathcal{Q}_n(X) \rightarrow Y$. In particular, the number of distinguishable states in Y cannot be less than that in $\mathcal{Q}_n(X)$. Thus, $\mathcal{Q}_n(X)$ is essentially the “smallest” model of X that still has the same aspect-essential properties.

Proof Sketch: The monotonicity $I(\mathcal{Q}_n(X)) \leq I(X)$ is a direct consequence of each closure being a quotient. When we apply a closure $\text{varsigma}_i: X \rightarrow \text{varsigma}_i(X)$, states of X are identified (merged), and no new distinctions are introduced. The map $\eta_i: X \rightarrow \text{varsigma}_i(X)$ is injective or at least information-lossless in the sense of adding no information; if anything, it's surjective many-to-one, which can only decrease the entropy or number of states. Therefore $I(\text{varsigma}_i(X)) \leq I(X)$ for each i . By induction (or by repeated application), applying a series of closures $\text{varsigma}_n(X) \leq I(X)$, and clearly if any identification actually merged distinct states, some information is lost, $\text{varsigma}_{\{2\}}, \dots$ yields a chain $X \rightarrow \text{varsigma}_{\{1\}}(X) \rightarrow \text{varsigma}_{\{2\}}(\text{varsigma}_{\{1\}}(X)) \rightarrow \dots$, and at each step information does not increase. Thus $I(\mathcal{Q}_n(X)) \leq I(X)$ (so the inequality is strict unless no merging occurred) ³⁰. The optimality part follows from the universal property of the composite closure $\mathcal{Q}_n$. By construction, $\mathcal{Q}_n(X)$ is X factored by the finest equivalence relation that still respects all aspect invariants. Any other system Y that also respects those invariants can be thought of as factoring X by some coarser (or equal) equivalence relation – coarser because Y might identify even more states than $\mathcal{Q}_n(X)$ does, potentially losing some invariant information. But if Y does not lose any invariant info, that means its identifications are among those already permitted by $\mathcal{Q}_n(X)$. Formally, there is a unique factorization $X \twoheadrightarrow \mathcal{Q}_n(X) \twoheadrightarrow Y$. Thus $\mathcal{Q}_n(X)$, which has removed all redundant distinctions, will never have more states than Y (up to isomorphism). This argument establishes $\mathcal{Q}_n(X)$ as the **most compressed** system capturing X 's essential structure in those n dimensions ³¹.

Interpretation: Information monotonicity assures us that by adding closures (i.e. by being pickier about what counts as “the same”), we cannot accidentally gain information. We only identify states, which reduces entropy or leaves it unchanged. In effect, each closure is a kind of **lossless compression** along its dimension: it discards only *redundant* information (differences that don't matter for that aspect), and keeps everything that does matter. The combined closure $\mathcal{Q}_n$ is then a compression that keeps all information required for *any* of the n aspects. If you know the state of $\mathcal{Q}_n(X)$, you know everything about the original state that could be known through algebraic queries, geometric proximity,

spectral modes, logical predicates, symmetries, causal relationships, context responses, topological features, or trajectory outcomes. This is why we call $\mathcal{Q}_n(X)$ a “sufficient” representation – nothing important to those aspects has been lost. The theorem’s optimality clause says it’s also the **minimal** such representation: any further compression would inevitably lose some information about at least one aspect.

It is also worth noting a dual perspective: the extended closure framework provides a powerful way to do **problem solving by temporarily increasing dimensionality**, confident that we can later compress back down without losing the insights gained. This might sound counterintuitive (since closures remove info, not add), but consider that by introducing new structures like symmetry parameters, contextual variables, or considering the entire trajectory space, we might transform a problem into a higher-dimensional one where its structure becomes simpler or more tractable. For example, a complex system might be easier to analyze if we **lift** it to a representation that makes a symmetry explicit (adding dimensions for group parameters can linearize certain dynamics), or if we embed the state space into a space of paths (turning a complicated state constraint into a simpler path constraint). In such a scenario, one could work in the expanded space to derive a solution or understanding, and then use the $\varSigma$ closures to **project back down** to the original space in a principled way. Because $\mathcal{Q}_n$ is *idempotent and lossless for invariant information, doing this “round trip” (expanding and then quotienting back) does not lose any information that was relevant – it only ensures that when we return to the original dimensionality, we ignore any artifacts of the expansion that aren’t truly meaningful. As a concrete illustration, imagine taking a one-dimensional time series and embedding it into a higher-dimensional phase space with added contextual variables and symmetry coordinates; in that space, certain patterns or cycles might become obvious. After analyzing or solving a task in the higher space, the closures $\varSigma, \dots$ can be applied to collapse the solution back to an equivalent one in the original 1D system. The extended framework guarantees that this collapsing is valid and optimal. In summary, $\varSigma_{\text{ctx}}, \varSigma_{\text{coh}}$ **the closures provide a reversible abstraction mechanism**: they allow us to compress a system optimally, but also play nicely with any temporary *decompression* (augmentation of the system) we might do for analytical convenience, since we know exactly how to quotient out the auxiliary structure afterward using the same closures.*

5.4 Discussion – Synthesis of Higher-Dimensional Closures

Bringing together all the above results, we can appreciate the expanded KUSMAN $\mathcal{Q}$ - $\varSigma$ framework as a powerful, unified approach to model reduction across multiple domains of system structure:

All-Aspects-at-Once Reduction: We have shown that nine different aspects of a system – spanning dynamic behavior, geometry, timescales, observations, symmetry transformations, causal structure, environmental context, topology, and entire trajectories – can all be incorporated into a single reduction pipeline. One can literally apply **all of these closure operators simultaneously** (thanks to their mutual commutativity) and end up with a coherent quotient system. This means the system can be **simplified along all these dimensions at once**, with the extended global aggregator $\mathcal{Q}_n$ yielding a version of X that has *no redundant degrees of freedom in any of those aspects*. This is the ultimate form of cross-domain simplification: instead of tackling different simplification goals separately, we enforce all invariants in parallel and still arrive at a well-defined result. The fact that $\mathcal{Q}_n$ remains idempotent and well-behaved assures us that this all-in-one reduction is mathematically sound. In practical terms, $\mathcal{Q}_n(X)$ can be seen as the **“information core”** of the system X with respect to everything we care about. It’s a sort of canonical form that strips away all inessential details at once.

Cross-Domain Compatibility (Commuting Invariants): A striking insight from our work is that invariants from very different domains can coexist without interfering. Initially, one might worry that imposing, say, a symmetry reduction and a causal reduction together could conflict – perhaps identifying certain states under symmetry could spoil a causal distinction or vice versa. However, our commutativity proofs show that **if something is an invariant in one sense and something else is an invariant in another sense, both can be imposed**. They are logically compatible, provided both were valid aspects of the original system. For example, preserving topological loops does not hinder merging states that are behaviorally equivalent, and preserving observational equivalences does not conflict with factoring out symmetries. This **independence (or orthogonality) of aspects** is not obvious a priori, but we’ve demonstrated it in each case. It suggests a deep principle: the closures we defined target fundamentally independent facets of structure, so that **simplifying one facet does not create “false distinctions” or erase true distinctions in another facet** ³². They truly commute. In categorical terms, each closure functor lives in its own “layer” but interacts via natural transformations that turn out to be isomorphisms, not merely one-way inequalities. From a broader perspective, this cross-domain commutativity underscores a form of *universality* of our approach – it doesn’t matter whether an invariant comes from physics, logic, topology, etc., as long as it is well-defined on the $\mathcal{MMK}$ structure, it can join the party of reductions and will play nicely with the others. This answers a key question: **Can we integrate new domain-specific simplifications into the framework without breaking it?** The answer is yes, as we’ve shown with five examples (and one can imagine adding more in the future).

Categorical Foundation vs. Pointwise Perspective: We navigated between two viewpoints: (1) the **categorical** (global) view, where each σ_i is a functor (specifically, a reflector that is left adjoint to the inclusion of a subcategory of $\mathcal{MMK}$ consisting of σ_i -closed systems) and (2) the **pointwise** (equivalence relation) view, where each closure corresponds to “gluing together” certain states of a single system. Initially, we defined and proved properties of closures using the categorical approach to ensure things like existence, uniqueness, and functoriality of the quotients. However, once commutativity is established, the pointwise view becomes very clear: since order doesn’t matter, we can imagine all equivalence relations being applied at once — effectively the intersection of all nine equivalences is itself an equivalence relation capturing all aspects at once. Therefore, $\mathcal{Q}_n(X)$ *can be described concretely as X modulo the relation “ x_1 and x_2 are indistinguishable by any of the aspect observations/transformations.” This is a single relation that is the **join** of the individual ones. In fact, one could have defined that relation first (“ $\sim = \sim_a \wedge \sim_g \wedge \dots \wedge \sim_n$ ” is well-defined. The $\sim$) and then shown it yields the same quotient as composing the closures one by one – that is another way to see why **equivalence of these perspectives** vindicates the idea that the category theory wasn’t an abstract indulgence but a rigorous way to guarantee that our intuitive “merge all indistinguishable states” procedure works out. For the readers already familiar with KUSMAN $\mathcal{Q}$ - ζ closure theory, this extended result confirms that the nice properties (functoriality, natural transformations commuting, etc.) still hold in the extended context, and that one can safely think of $\mathcal{Q}_n(X)$ as simply a big quotient by all the invariants at once (with the formal guarantee that this quotient exists and is unique up to isomorphism).*

Unified Principle of Invariant-Based Reduction: Perhaps the most profound conclusion of Stage 5 is the realization that *any* meaningful notion of “irrelevant difference” can be incorporated into the framework. Each closure operator we added corresponds to an equivalence relation derived from a distinct concept of invariance (symmetry invariance, causal irrelevance, context independence, topological invariance, path equivalence). What we have seen is that **all such equivalences, as long as they respect the basic structure (metrics, measures, kernels), become just another idempotent closure in $\mathcal{MMK}$** . In other words, the framework is *infinitely extensible*: if tomorrow we decide that some new aspect (say,

“energy landscape connectivity” or “functional equivalence in a neural network”) is important and we can formalize an equivalence relation for it, then we can add σ_{new} and expect it to commute with the others, because it’s fundamentally the same kind of thing – a quotient by a system invariant. This ties back to a categorical insight: the closures we have are all examples of **reflective subcategories** of $\mathcal{MMK}$ (each defined by some invariant property), and the intersection of reflective subcategories is reflective as well (under appropriate conditions) – which is why their intersection (the combined quotient) exists and is reflective. Although we won’t go deeper into category theory here, it’s worth noting that in the original KUSMAN work the four core closures were related to sheaf-theoretic *Lawvere–Tierney topologies* on an internal topos (essentially each closure was like “forcing” a certain kind of axiom to become true, akin to sheafification) ³³. Our extended closures likely have a similar interpretation (for instance, symmetry closure is a quotient by a group action – a well-known universal construction; causal closure might relate to a certain sigma-algebra on path space; contextual closure to a fibred product or averaging functor; etc.). The **Unified Dimensionality Theorem** we proved is, in a sense, a guarantee that all these arise from a single **universal principle**: identify and quotient out invariant structure. And because they all arise from the same principle, they are *compatible*. This gives confidence that the KUSMAN Φ - ζ framework can serve as a **universal platform for complexity reduction** across disciplines.

In summary, Stage 5 has achieved a comprehensive generalization of the closure-based reduction framework. We added five new closure operators to capture symmetry, causal, contextual, topological, and trajectory-level aspects of systems, and we rigorously demonstrated that **they integrate seamlessly** with the original four. The extended global reducer $\mathcal{Q}_{\text{ext}} = \mathcal{Q}_n$ remains idempotent and produces an optimally simplified model of the system, one that **simultaneously preserves all specified invariants**. All the closures commute pairwise, confirming that reductions along different dimensions of structure can be performed together without issue. This leads to a rich lattice of intermediate models and a guarantee of information monotonicity – adding more reduction criteria never increases complexity. The end result is a highly flexible yet mathematically disciplined framework for model simplification. It tells us that **all meaningful simplifications of a dynamical system can be viewed as factoring out an equivalence relation corresponding to some invariant**, and crucially, **all such simplifications commute when properly formalized**. This unification not only provides a powerful theoretical tool but also offers practical guidance: whether one is dealing with a physical system with symmetries, a biological system with context-dependent behaviors, or a computational process with irrelevant steps, one can plug the relevant closures into $\mathcal{Q}_n$ and obtain a canonical reduced model. The framework thus stands as a proof-of-concept that *we can tame complexity in a principled, multi-dimensional way*, confirming the vision that motivated the KUSMAN Φ - ζ approach.

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